

SHI-XIN ZHANG (张士欣)

✉ znfesnpbh@gmail.com · 🌐 refraction-ray · 📄 re-ra.xyz

Associate Professor at IOP CAS and former Senior Research Scientist at Tencent. Pioneering the intersection of Quantum Computing and Artificial Intelligence. Creator of TensorCircuit, a world-leading open-source quantum framework. Distinguished for bridging fundamental physics insights with robust engineering practices to deliver advanced tools and solutions, backed by 30+ patents and 50+ publications.

EXPERIENCE

Institute of Physics, Chinese Academy of Sciences Associate Professor 2024 –

- Principal Investigator

Established and leading the research group focusing on quantum physics and computation. Mentoring a multi-disciplinary team of PhD students, postdocs and undergraduate students to bridge the gap between deep learning infrastructure and quantum many-body physics. Secured competitive research grants (NSFC/CAS/National Science and Technology Major Project).

Tencent Quantum Laboratory Senior Research Scientist 2021 – 2024

- Quantum algorithms and applications: Project Owner / Research Scientist

Led a small group of talented people focusing on the research, analysis, and design for near term quantum algorithms and quantum simulation schemes. Also collaborated with top commercial partners from the finance/biology/energy/material/IT sectors, exploring industry solutions with potential quantum advantage.

- Quantum Software: Software Creator / Platform Architect / Core Author and Maintainer

Created and developed a high performance open-source and full-featured quantum software framework: TensorCircuit. The software is empowered by an advanced tensor network engine, and directly built on top of machine learning frameworks, enabling next-generation quantum computing infrastructure.

EDUCATION

Tsinghua University PhD in Physics 2016 – 2021

Institute for Advanced Study Advisor: Prof. Hong Yao

PhD Thesis: Differentiable Programming in Quantum Physics

Outstanding PhD Award (87 amongst 2981) / Outstanding PhD thesis at Tsinghua University

Tsinghua University BSc in Physics 2012 – 2016

Department of Physics GPA: 95/100 Rank: 1/95

Top 1 in 2012 National College Entrance Examination in Hebei Province

RESEARCH

Interests

- Quantum computing: quantum algorithm, quantum simulation, quantum noise, quantum software, quantum information.
- Artificial intelligence: the interplay between machine learning infrastructure/models/philosophy with quantum many-body physics, quantum machine learning.
- Non-equilibrium physics: novel phases and phase transitions in non-equilibrium quantum systems including quantum chaos, many-body localization, time crystal, (monitored) random quantum circuits, etc.

Publications

63 publications and preprints in total (52 as the first or corresponding author), 48 published in peer-reviewed journals (40 as the first or corresponding author), including 9 in Physical Review Letters (2 as the first author and 5 as the corresponding author) and 1 in Nature Communications (1 as the corresponding author). Selected works as the **first**, **joint first**[†] and **corresponding**^{*} author are listed below, please see my Google Scholar profile for the full publication list.

1. Yu-Qin Chen*, and **Shi-Xin Zhang***, *Intrinsic preservation of plasticity in continual quantum learning*, PRX Quantum **7**, 033003 (2025).
2. Chun-Yue Zhang, Zi-Xiang Li*, and **Shi-Xin Zhang***, *Entanglement Growth from Entangled States: A Unified Perspective on Entanglement Generation and Transport*, Physical Review Letters **137**, 020404 (2026).
3. Wei-Xuan Chang, Shuai Yin*, **Shi-Xin Zhang***, and Zi-Xiang Li*, *Imaginary-time Mpemba effect in quantum many-body systems*, Physical Review Letters **136**, 100403 (2026).
4. Shuo Liu[†], Hao-Kai Zhang[†], Shuai Yin, and **Shi-Xin Zhang***, *Symmetry restoration and quantum Mpemba effect in symmetric random circuits*, Physical Review Letters **133**, 140405 (2024).
5. Shuo Liu, Ming-Rui Li, **Shi-Xin Zhang***, and Shao-Kai Jian*, *Entanglement structure and information protection in noisy hybrid quantum circuits*, Physical Review Letters, **132**, 240402 (2024).
6. Shuo Liu, **Shi-Xin Zhang***, Chang-Yu Hsieh, Shengyu Zhang, and Hong Yao*, *Discrete time crystal enabled by Stark many-body localization*, Physical Review Letters **130**, 120403 (2023).
7. **Shi-Xin Zhang[†]**, Zhou-Quan Wan[†], Chee-Kong Lee, Chang-Yu Hsieh*, Shengyu Zhang, and Hong Yao*, *Variational quantum-neural hybrid eigensolver*, Physical Review Letters **128**, 120502 (2022).
8. **Shi-Xin Zhang** and Hong Yao*, *Universal properties of many-body localization transitions in quasiperiodic systems*, Physical Review Letters **121**, 206601 (2018).
9. Yu-Qin Chen* and **Shi-Xin Zhang***, *Superior resilience to poisoning and amenability to unlearning in quantum machine learning*, Nature Communications **17**, 3716 (2026).
10. Shuo Liu, Shao-Kai Jian*, and **Shi-Xin Zhang***, *Noisy Monitored Quantum Circuits*, Journal of Physics: Condensed Matter **38**, 203001 (2026).
11. Han-Ze Li, Ching Hua Lee, Shuo Liu*, **Shi-Xin Zhang***, and Jian-Xin Zhong*, *Quantum Mpemba effect in long-ranged $U(1)$ -symmetric random circuits*, Physical Review B **113**, 134310 (2026).
12. Hui Yu, Jiangping Hu*, and **Shi-Xin Zhang***, *Quantum Pontus-Mpemba Effects in Real and Imaginary-time Dynamics*, Physical Review B **113**, 134304 (2026).
13. Zong-Liang Li, and **Shi-Xin Zhang***, *The Dual Role of Low-Weight Pauli Propagation: A Flawed Simulator but a Powerful Initializer for Variational Quantum Algorithms*, Physical Review Research **8**, 013266(2026).
14. Yuqi Qing, Yu-Qin Chen*, **Shi-Xin Zhang***, *Entanglement growth and information capacity in a quasiperiodic system with a single-particle mobility edge*, Physical Review B **113**, 064308 (2026).
15. Yu-Qin Chen*, **Shi-Xin Zhang[†]**, and Shengyu Zhang*, *Non-Markovianity benefits quantum dynamics simulation*, Advanced Quantum Technologies **9**, e00405 (2026).
16. Shuo Liu[†], Hao-Kai Zhang[†], Shuai Yin, **Shi-Xin Zhang*** and Hong Yao*, *Symmetry restoration and quantum Mpemba effects in many-body localization systems*, Science Bulletin **70**, 3991 (2025).
17. Sibao Guo, Shuai Yin, **Shi-Xin Zhang***, and Zi-Xiang Li*, *Skin Effect Induced Anomalous Dynamics from Charge-Fluctuating Initial States*, Physical Review B **112**, 155419 (2025).
18. Hui Yu, Zi-Xiang Li*, and **Shi-Xin Zhang***, *Symmetry breaking dynamics in quantum many-body systems*, Chinese Physics Letters **42**, 110602 (2025).
19. Yu-Qin Chen* and **Shi-Xin Zhang***, *Effective temperature in approximate quantum many-body states*, Physical Review Research **7**, 033185 (2025).
20. Shuo Liu[†], Zhengzhi Wu[†], **Shi-Xin Zhang***, and Hong Yao*, *Supersymmetry dynamics on Rydberg atom arrays*, Physical Review B **112**, L020301 (2025).
21. Hui Yu, Shuo Liu, and **Shi-Xin Zhang***, *Quantum Mpemba effects from symmetry perspectives*, AAPS Bulletin **35**, 17 (2025).
22. Xu Feng, Shuo Liu, **Shi-Xin Zhang***, and Shu Chen*, *Numerical instability of non-Hermitian Hamiltonian evolution*, Physical Review B **111**, 224310 (2025).
23. Yu-Qin Chen, Shuo Liu, and **Shi-Xin Zhang***, *Subsystem information capacity in random circuits and Hamiltonian dynamics*, Quantum **9**, 1783 (2025).
24. **Shi-Xin Zhang[†]**, Jiaqi Miao[†], and Chang-Yu Hsieh*, *Variational post-selection for ground states and thermal states simulation*, Quantum Science and Technology **10**, 015028 (2025).
25. Yi-Ming Ding*, Yan-Cheng Wang, **Shi-Xin Zhang***, and Zheng Yan*, *Exploring the topological sector optimization on quantum computers*, Physical Review Applied **22**, 034031 (2024).
26. Shuo Liu, Ming-Rui Li, **Shi-Xin Zhang***, Shao-Kai Jian*, and Hong Yao* *Noise-induced phase transitions in hybrid quantum circuits*, Physical Review B **110**, 064323 (2024).

27. **Shi-Xin Zhang**^{*} and Shuai Yin^{*}, *Universal imaginary-time critical dynamics on a quantum computer*, Physical Review B **109**, 134309 (2024).
28. Lixue Cheng[†], Yu-Qin Chen[†], **Shi-Xin Zhang**^{*}, and Shengyu Zhang^{*}, *Quantum approximate optimization via learning-based adaptive optimization*, Communications Physics **7**, 83 (2024).
29. Jiaqi Miao, Chang-Yu Hsieh^{*}, and **Shi-Xin Zhang**^{*}, *Neural network encoded variational quantum algorithms*, Physical Review Applied **21**, 014053 (2024).
30. Shuo Liu, **Shi-Xin Zhang**^{*}, Shao-Kai Jian^{*}, and Hong Yao^{*}, *Training variational quantum algorithms with random gate activation*, Physical Review Research **5**, L032040 (2023).
31. **Shi-Xin Zhang**, Zhou-Quan Wan, Chang-Yu Hsieh^{*}, Hong Yao^{*}, and Shengyu Zhang^{*}, *Variational quantum-neural hybrid error mitigation*, Advanced Quantum Technologies, **6**, 2300147 (2023).
32. **Shi-Xin Zhang**[†], Zhou-Quan Wan[†], and Hong Yao^{*}, *Automatic differentiable Monte Carlo: theory and application*, Physical Review Research **5**, 033041 (2023).
33. Shuo Liu[†], Ming-Rui Li[†], **Shi-Xin Zhang**^{*}, Shao-Kai Jian^{*}, and Hong Yao^{*}, *Universal Kardar-Parisi-Zhang scaling in noisy hybrid quantum circuits*, Physical Review B **107**, L201113 (2023).
34. **Shi-Xin Zhang**, et al., *TensorCircuit: a quantum software framework for the NISQ era*, Quantum **7**, 912 (2023).
35. Shuo Liu, **Shi-Xin Zhang**^{*}, Chang-Yu Hsieh^{*}, Shengyu Zhang, and Hong Yao^{*}, *Probing many-body localization by excited-state variational quantum eigensolver*, Physical Review B **107**, 024204 (2023).
36. **Shi-Xin Zhang**, Chang-Yu Hsieh^{*}, Shengyu Zhang, and Hong Yao^{*}, *Differentiable quantum architecture search*, Quantum Science and Technology **7**, 045023 (2022).
37. Zhou-Quan Wan, **Shi-Xin Zhang**^{*}, and Hong Yao^{*}, *Mitigating the fermion sign problem by automatic differentiation*, Physical Review B **106**, L241109 (2022).
38. **Shi-Xin Zhang**, Chang-Yu Hsieh^{*}, Shengyu Zhang, and Hong Yao^{*}, *Neural predictor based quantum architecture search*, Machine Learning: Science and Technology **2**, 045027 (2021).
39. **Shi-Xin Zhang**, Shao-Kai Jian, and Hong Yao^{*}, *Quantum criticality preempted by nematicity*, Physical Review B **103**, 165129 (2021).
40. **Shi-Xin Zhang**, Shao-Kai Jian, and Hong Yao^{*}, *Correlated triple-Weyl semimetals with Coulomb interactions*, Physical Review B (Rapid Communication) **96**, 241111 (2017).
41. **Shi-Xin Zhang**^{*} and Yu-Qin Chen^{*}, *ORBIT-Q: Dual-axis benchmarking of autonomous agents in scientific quantum programming*, arXiv:2607.03105 (2026).
42. **Shi-Xin Zhang**^{*} and Yu-Qin Chen^{*}, *Quantum Subliminal Learning*, arXiv:2605.29557 (2026).
43. Hui Yu, Jiangping Hu^{*}, and **Shi-Xin Zhang**^{*}, *Post-Selection-Free Decoding of Measurement-Induced Area-Law Phases via Neural Networks*, arXiv:2604.03550 (2026).
44. **Shi-Xin Zhang**^{*}, Yu-Qin Chen, Weitang Li, Jiace Sun, Wei-Guo Ma, Pei-Lin Zheng, Yu-Xiang Huang, Qi-Xiang Wang, Hui Yu, Zhuo Li, Xuyang Huang, Zong-Liang Li, Zhou-Quan Wan, Shuo Liu, Jiezhong Qiu, Jiaqi Miao, Zixuan Song, Yuxuan Yan, Kazuki Tsuoka, Pan Zhang, Lei Wang, Heng Fan, Chang-Yu Hsieh, Hong Yao, and Tao Xiang, *TensorCircuit-NG: A Universal, Composable, and Scalable Platform for Quantum Computing and Quantum Simulation*, arXiv:2602.14167 (2026).
45. Chun-Yue Zhang, **Shi-Xin Zhang**^{*}, and Zi-Xiang Li^{*}, *Bond Additivity and Persistent Geometric Imprints of Entanglement in Quantum Thermalization*, arXiv:2601.01327 (2026).
46. Xu Feng, Shuo Liu, Shu Chen^{*}, and **Shi-Xin Zhang**^{*}, *Tunable discrete quasi-time crystal from a single drive*, arXiv:2512.10303 (2025).
47. Wei-Guo Ma, Heng Fan^{*}, and **Shi-Xin Zhang**^{*}, *A Qudit-native Framework for Discrete Time Crystals*, arXiv:2512.04577 (2025).
48. Hui Yu, Jiangping Hu^{*}, and **Shi-Xin Zhang**^{*}, *Hilbert subspace imprint: a new mechanism for non-thermalization*, arXiv:2506.11922 (2025).
49. **Shi-Xin Zhang**^{*}, and Lei Wang^{*}, *A Unified Variational Framework for Quantum Excited States*, arXiv:2504.21459 (2025).
50. **Shi-Xin Zhang**, *Classification on the computational complexity of spin models*, arXiv:1911.04122 (2019).
51. Zhou-Quan Wan and **Shi-Xin Zhang**^{*}, *Automatic differentiation for complex valued SVD*, arXiv:1909.02659 (2019).
52. **Shi-Xin Zhang** and Hong Yao^{*}, *Strong and weak many-body localizations*, arXiv:1906.00971 (2019).

Patents

40 domestic and international patent applications (24 authorized) in the fields of quantum circuit design automation, quantum AI hybrid solutions, quantum software acceleration, and quantum simulation.

HONORS

- At Tsinghua University, won awards including National Scholarship, National Encouragement Scholarship, Future Scholar Scholarship, First Class Freshmen Scholarship, Zhang Mingwei Scholarship, Xuetao Talent Program Scholarship, etc.
- At Tencent, rated as outstanding (10%) and selected as the outstanding individual of the lab. Also won the Tencent Outstanding R&D award for development of the quantum computing platform and the Tencent Outstanding Course Award for variational quantum computing lectures.

SKILLS

- The interplay between quantum physics and computer science: Familiar with quantum computation, quantum artificial intelligence, and machine learning in quantum physics.
- Condensed matter physics: Familiar with the basic theory and methods for quantum many-body physics. Know about numerical methods including tensor network, quantum Monte Carlo, mean field, variational approach, and exact diagonalization.
- High-performance computation: Built the full-stack cluster in IASTU. Familiar with toolchains and the ecosystem in Ops, HPC, GPU, and cloud computation.
- DevOps for science: Applying CI/CD, unit testing, continuous integration, and modular design to scientific research codes to ensure reproducibility and extensibility.
- Python: Familiar with Python language and third-party packages for scientific computing, data science, machine learning, web development, web crawling, software engineering, etc.
- Differentiable programming, probabilistic programming and quantum programming: Familiar with the programming paradigm and ecosystem: TensorFlow, Jax, PyTorch, Keras, TensorNetwork, Qiskit, Cirq, TensorFlow Quantum, PennyLane, Mitiq, etc.
- Programming language: Python, Mathematica, C++, Julia, JavaScript, Bash; Markup language: HTML, CSS, Markdown, reStructuredText, $\LaTeX$; Natural language: Chinese, English, Korean.

OPEN SOURCE CONTRIBUTIONS

Familiar with the open source practice and created many popular open-source projects and platforms related to HPC, scientific computing, finance, and AI with **3000+** stars/forks and 300 followers. Also contributed to several famous open-source projects including NumPy, TensorFlow, Autograd, TensorNetwork, TensorFlow Quantum, and conda-smithy. Please refer to my GitHub Profile for details.

Projects

- TensorCircuit & TensorCircuit-NG: Full-featured, high-performance quantum software framework designed for the NISQ era. The package receives **1M+** downloads and 180+ academic citations. TensorCircuit is top 10 events for quantum industry in 2022 by QuantumChina, recommended quantum software in Google Summer of Code 2023, and hackathon projects in Unitary Hack 2024 and summer-ospp 2025.
- TensorNetwork-NG: This is a continuously maintained fork of the popular tensor network package, originally released by Google and designed for the machine learning era.
- qop: The software supports algebra on complex number, quaternion, boson, hardcore boson, fermion, spin, and einsum on symbols.
- realspace-RG: Numerical implementation of renormalization group on many-body localization phase transition. Highly parallelized C++ code used on Tianhe II Supercomputer.
- xalpha: Analysis, management and backtest on financial investment. The software has **2300+** stars with a relatively large open-source community.
- hpc-ansible: This toolset provides all components required to build the supercomputer from bootstrap. With vm-cluster toolset, users can build a KVM cluster in one click.

- subway: The software enables HPC job management in an automatic fashion, by providing a highly customized pipeline from data management to job submission.
- myarxiv-app: The project builds the full-stack web for arXiv papers, utilizing techniques including Vue, Webpack, Flask, Docker Compose, web crawler, and NLP algorithms.